

DANDI:001417: Allen Institute OpenScope - Psycode

Region assignment (computed by this export): each unit lists all channels of its probe. In 2 of the 20 files the peak channel is given (units.extremum_channel_index) and was used. The other 18 have no peak-channel column, so there the region is the electrode location nearest to units.depth / units.estimated_x. On a file with both, that method agreed with the peak channel for 95.6% of 3,287 units.

Scope of this export (computed by this export): the units, electrodes and intervals tables of every NWB file were read remotely from the DANDI Archive; voltage, LFP, video and other time series were not downloaded.

Source labels: **DANDI metadata** = the Dandiset's published metadata; **NWB file** = values or text stored in the NWB files; **Computed by this export** = derived by the export scripts in dandi_export/ (definitions given where used).

1. The dataset as published

Everything in this section is quoted from the Dandiset's published metadata or, for file-level fields, from sub-752309_ses-ecephys-752309-2025-02-11-14-00-40_ecephys.nwb.

Field	Value (DANDI metadata)
Name	Allen Institute OpenScope - Psycode
Identifier / version	DANDI:001417 / 0.260626.2154
Citation	Amaya, Avalon; Bawany, Ahad; Buice, Michael; Carlson, Mikayla; De Filippo, Roberto; Durand, Severine; Gillis, Ryan; Grasso, Conor; Groblewski, Peter; Han, Warren; Johnson, Tye; Kenney, Jaimie; Kiselycznyk, Carly; Koch, Christof; Lecoq, Jérôme A.; Loeffler, Henry; Marks, Lydia; Naidoo, Robyn; Ott, Torben; Ouellette, Ben; Peene, Ross Carter; Rembado, Irene; Schmitz, Dietmar; Suarez, Lucas; Swapp, Jackie; Weber, Julie; Wilkes, Josh; Williford, Ali; Wyrick, David (2026) Allen Institute OpenScope - Psycode (Version 0.260626.2154) [Data set]. DANDI Archive. https://doi.org/10.48324/dandi.001417/0.260626.2154
DOI	10.48324/dandi.001417/0.260626.2154
Contributors (roles)	National Institutes of Health (NIH) (Funder); Amaya, Avalon (Researcher); Bawany, Ahad (Software); Buice, Michael (Researcher); Carlson, Mikayla (DataCollector); De Filippo, Roberto (ContactPerson, Conceptualization); Durand, Severine (Investigation); Gillis, Ryan (DataCollector); Grasso, Conor (Researcher); Groblewski, Peter (Researcher); Han, Warren (Researcher); Johnson, Tye (Researcher); Kenney, Jaimie (Researcher); Kiselycznyk, Carly (ProjectAdministration); Koch, Christof (Researcher); Lecoq, Jérôme A. (ProjectLeader, ContactPerson); Loeffler, Henry (Researcher); Marks, Lydia (Researcher); Naidoo, Robyn (Researcher); Ott, Torben (Supervision); Ouellette, Ben (Researcher); Peene, Ross Carter (Software); Rembado, Irene (Supervision); Schmitz, Dietmar (Conceptualization); Suarez, Lucas (Researcher); Swapp, Jackie (Researcher); Weber, Julie (Researcher); Wilkes, Josh (Researcher); Williford, Ali (Researcher); Wyrick, David (FormalAnalysis)
License	CC-BY-4.0
Keywords	psilocybin, openscope, mouse, vision, ecephys, neuropixels
Study target	Psilocybin; Mouse; Vision
Ethics approval	Allen Institute Institutional Animal Care and Use Committee (IACUC), protocol 2117
Related resources	OpenScope's Psycode Dataset https://alleninstitute.github.io/openscope_databook/psycode/ (IsDescribedBy)
Anatomy (about)	Psilocybin
Files / size / subjects	20 files, 354.66 GB, 10 subjects
Species	Mus musculus - House mouse
Approach	electrophysiological approach
Measurement technique	multi electrode extracellular electrophysiology recording technique; spike sorting technique; surgical technique; signal filtering technique; analytical technique
Variables measured	ProcessingModule, Units, LFP, ElectricalSeries, ElectrodeGroup
Date published	2026-06-26T21:54:04.988117+00:00

Description (DANDI metadata, verbatim)

Psycode dataset is an OpenScope project designed to investigate the effects of psilocybin on visual processing and behavior in the mouse visual cortex. The core experimental design follows a 'pre-drug / post-drug' structure, allowing for a within-subject comparison of neural activity and task performance. Initially, mice are trained on a visual change detection task, similar to the paradigm

used in the Visual Behavior Optical Physiology and Visual Behavior Neuropixels projects. In this go/no-go task, mice are shown a continuous series of briefly presented natural images (presented for 250 ms), and they earn water rewards by correctly reporting when the identity of the image changes. Alongside this behavior, the experiment includes a battery of passive visual stimulation protocols to probe the functional properties of visual neurons. The experimental timeline follows a structured sequence. Before psilocybin application, mice first undergo passive visual stimulation for receptive field (RF) mapping and a surround suppression paradigm, providing a baseline measurement of each neuron’s tuning properties. This is followed by the change detection task to end the pre-injection epoch. After psilocybin is administered, the same sequence of stimuli is presented in reverse: first the change detection task, followed by receptive field mapping and surround suppression. Finally, passive visual stimulation is repeated, now including an optotagging protocol at the end to identify cortical SST neurons. This symmetrical design—with both behavioral and passive protocols repeated before and after drug administration—enables a powerful within-subject analysis of how psilocybin modulates visual processing, from basic receptive field properties to complex task-dependent dynamics involving natural image perception. The dataset therefore offers a unique opportunity to link molecularly-specific cell types, natural image perception, and the state-dependent changes induced by a psychedelic compound. **Note: In this dandiset, session ecephys-752309-2025-02-12-13-53-52 had synchronization/alignment issues between stimulus and ecephys timing. Session ecephys_767926_2025-02-26_15-49-35 is missing running data because of a problem ith the sensor during recording**

File-level fields (NWB file, verbatim)

Field	Value (NWB file)
session_description	NWB file generated by AIND pipeline
institution	Allen Institute for Neural Dynamics
subject	subject_id: 752309; species: Mus musculus; genotype: Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt; sex: F; age: P220D
devices	Device: Ecephys probe. Automatically generated.; ProbeA: Model: - Serial number: 20097906352; ProbeB: Model: - Serial number: 20097906092; ProbeC: Model: - Serial number: 20097909172; ProbeD: Model: - Serial number: 20097906681; ProbeE: Model: - Serial number: 20097906241; ProbeF: Model: - Serial number: 17131307182
electrode_groups	ProbeA: description = 'no description', location = 'unknown'; ProbeB: description = 'no description', location = 'unknown'; ProbeC: description = 'no description', location = 'unknown'; ProbeD: description = 'no description', location = 'unknown'; ProbeE: description = 'no description', location = 'unknown'; ProbeF: description = 'no description', location = 'unknown'

2. Contents of the files

Quantity	Value	Source
NWB files read / subjects	20 / 10	NWB file
Species	Mus musculus (20)	NWB file
Sex (files)	F (14), M (6)	NWB file
Genotype (files)	Sst-IRES-Cre/wt;Ai32(RCL-ChR2(H134R)_EYFP)/wt (20)	NWB file
Strain (files)		NWB file
Age (files)	P201D (2), P220D (1), P221D (1), P200D (1), P208D (1), P209D (1), P202D (1), P175D (1), P176D (1), P189D (1), P190D (1), P144D (1), P145D (1), P157D (1), P158D (1), P184D (1), P185D (1), P163D (1), P164D (1)	NWB file
Lab / institution	/ Allen Institute for Neural Dynamics (20)	NWB file
Units (total)	82,742	NWB file
Units per file	median 4100 (range 3287–5057)	NWB file
Spikes (sum of spike_times_index lengths)		NWB file
How region was assigned to units	electrodes.location at the electrode (among the unit's units.electrodes) nearest to units.depth/estimated_x (file has no peak-channel column); electrodes.location at units.electrodes[extremum_channel_index] (each unit lists all channels of its probe; position checked against units.depth)	Computed by this export
Distinct region labels	124	NWB file
units.default_qc values	True: 47,411, False: 35,331	NWB file
units.decoder_label values	sua: 32,104, mua: 30,918, noise: 19,720	NWB file

The meaning of each units column, including any quality label, is quoted from the NWB file in section 7.

3. NWB files

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-752309_ses-ecephys-752309-2025-02-11-14-00-40_ecephys	752309	F	P220D	17.87	4113	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(147); spontaneous_presentations(5099)
sub-752309_ses-ecephys-752309-2025-02-12-13-53-52_ecephys	752309	F	P221D	17.68	4055	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-752311_ses-ecephys-752311-2025-01-22-14-05-21_ecephys	752311	F	P200D	18.35	4331	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(149); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(151); spontaneous_presentations(5099)
sub-752311_ses-ecephys-752311-2025-01-23-14-00-04_ecephys	752311	F	P201D	18.04	4141	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-752312_ses-ecephys-752312-2025-01-30-11-44-04_ecephys	752312	F	P208D	14.93	3287	2400	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(147); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(148); spontaneous_presentations(5099)
sub-752312_ses-ecephys-752312-2025-01-31-11-02-20_ecephys	752312	F	P209D	18.62	4596	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-753316_ses-ecephys-753316-2025-01-28-13-57-06_ecephys	753316	F	P201D	17.96	4351	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(33); 5 hz pulse train_presentations(37); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(34); spontaneous_presentations(5099)

File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-753316_ses-ecephys-753316-2025-01-29-13-57-16_ecephys	753316	F	P202D	17.27	3854	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-760322_ses-ecephys-760322-2025-02-05-14-30-46_ecephys	760322	F	P175D	18.94	5057	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-760322_ses-ecephys-760322-2025-02-06-14-07-25_ecephys	760322	F	P176D	18.49	4743	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-760324_ses-ecephys-760324-2025-02-19-10-56-40_ecephys	760324	F	P189D	18.62	4626	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(150); behavior_presentations(3597); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5098)
sub-760324_ses-ecephys-760324-2025-02-20-11-31-01_ecephys	760324	F	P190D	18.24	4471	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-767925_ses-ecephys-767925-2025-02-13-11-40-19_ecephys	767925	M	P144D	17.66	3932	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(149); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-767925_ses-ecephys-767925-2025-02-14-11-04-33_ecephys	767925	M	P145D	17.20	3714	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(147); behavior_presentations(3597); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5098)

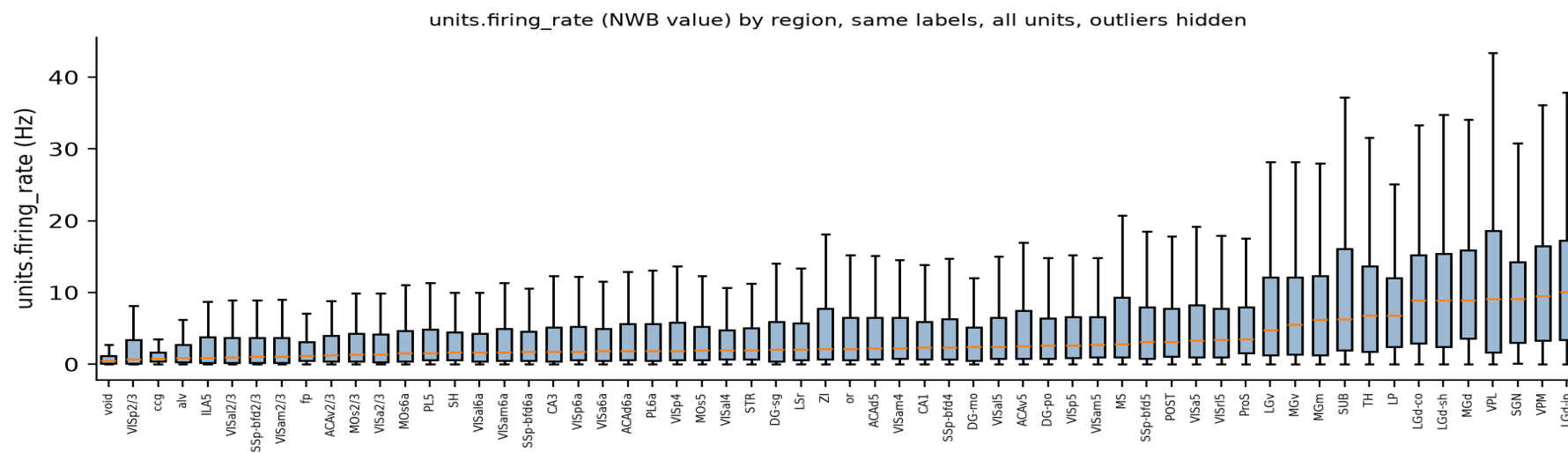
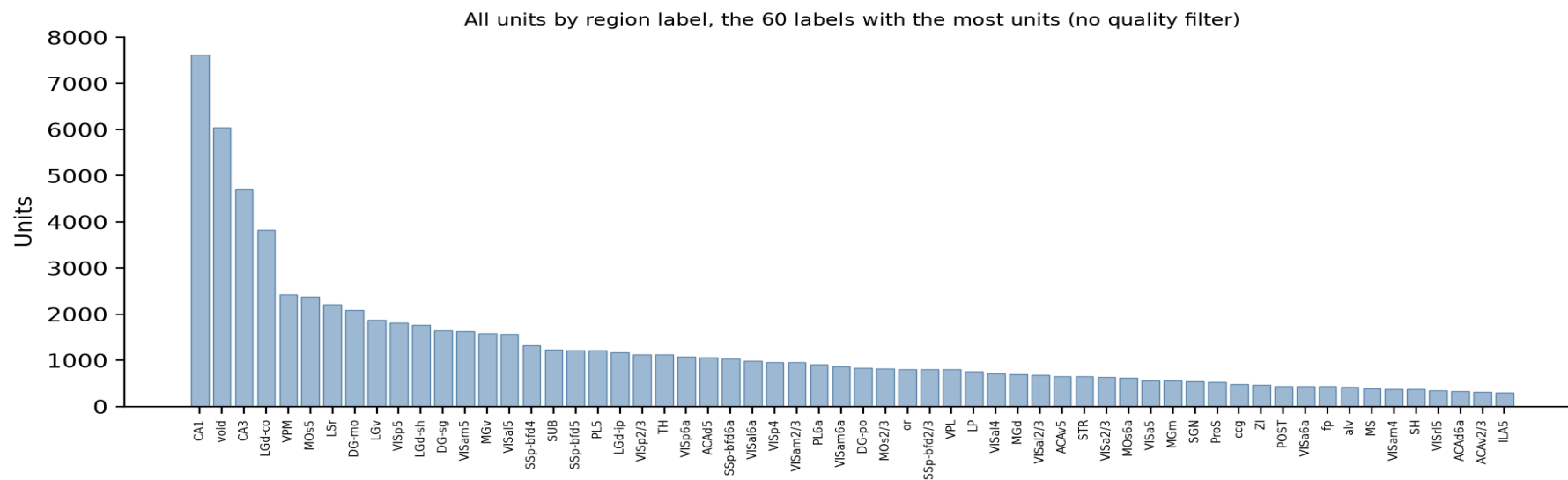
File (session_id)	Subject	Sex	Age	Size (GB)	Units	Electrodes	Electrode groups	Interval tables (rows)
sub-767926_ses-ecephys-767926-2025-02-26-15-49-35_ecephys	767926	M	P157D	18.97	4712	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-767926_ses-ecephys-767926-2025-02-27-15-12-01_ecephys	767926	M	P158D	17.75	4086	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5099)
sub-767932_ses-ecephys-767932-2025-03-25-14-35-45_ecephys	767932	F	P184D	17.53	3913	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(149); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(149); spontaneous_presentations(5099)
sub-767932_ses-ecephys-767932-2025-03-26-14-07-06_ecephys	767932	F	P185D	16.77	3443	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(150); behavior_presentations(3598); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(149); spontaneous_presentations(5099)
sub-773206_ses-ecephys-773206-2025-04-01-10-47-58_ecephys	773206	M	P163D	16.80	3558	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(150); 5 hz pulse train_presentations(149); behavior_presentations(3597); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5098)
sub-773206_ses-ecephys-773206-2025-04-02-11-46-05_ecephys	773206	M	P164D	17.00	3759	2880	ProbeA, ProbeB, ProbeC, ProbeD, ProbeE, ProbeF	40 hz pulse train_presentations(149); 5 hz pulse train_presentations(150); behavior_presentations(3597); post_receptive_field_mapping_presentations(1944); post_surround_suppression_mapping_presentations(750); pre_receptive_field_mapping_presentations(1944); pre_surround_suppression_mapping_presentations(750); raised_cosine_presentations(150); spontaneous_presentations(5098)

Other data in each file (not exported)

File (session_id)	Acquisition	Processing modules: interfaces
sub-752309_ses-ecephys-752309-2025-02-11-14-00-40_ecephys	autorewarded, licks, raw_running_wheel_rotation, reward_volume, running_wheel_signal_voltage, running_wheel_supply_voltage	ecephys:LFP; eye_tracking:likely_blink_times/corneal_reflection/ellipse/pupil; running:running_speed/running_wheel_rotation; stimulus:timestamps
sub-752309_ses-ecephys-752309-2025-02-12-13-53-52_ecephys	autorewarded, licks, raw_running_wheel_rotation, reward_volume, running_wheel_signal_voltage, running_wheel_supply_voltage	ecephys:LFP; eye_tracking:likely_blink_times/corneal_reflection/ellipse/pupil; running:running_speed/running_wheel_rotation; stimulus:timestamps

File (session_id)	Acquisition	Processing modules: interfaces
sub-773206_ses-ecephys-773206-2025-04-02-11-46-05_ecephys	autorewarded, licks, raw_running_wheel_rotation, reward_volume, running_wheel_signal_voltage, running_wheel_supply_voltage	ecephys:LFP; eye_tracking:likely_blink_times/corneal_reflection/ellipse/pupil; running:running_speed/running_wheel_rotation; stimulus:timestamps

4. Units per region (Computed by this export)



5. All region labels (Computed by this export)

One row per region label exactly as stored in the NWB files, all units (no quality filter), sorted by unit count. Firing-rate columns summarise the NWB units.firing_rate values where the dataset provides them.

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
CA1	7620	20	10	4.84	2.29
void	6047	20	10	1.01	0.48
CA3	4700	20	10	4.72	1.74
LGd-co	3825	20	10	10.35	8.88
VPM	2424	17	9	11.41	9.52
MOs5	2378	19	10	4.04	1.95
LSr	2216	15	9	4.16	2.06
DG-mo	2094	20	10	4.27	2.42
LGv	1879	20	10	8.41	4.76
VIsp5	1808	20	10	4.80	2.67
LGd-sh	1764	20	10	10.63	8.91
DG-sg	1644	20	10	4.83	2.00
VISam5	1630	20	10	4.82	2.75
MGv	1578	13	8	8.29	5.58
VISal5	1563	20	10	4.57	2.48
SSp-bfd4	1316	20	10	4.63	2.30
SUB	1224	15	9	10.44	6.28
SSp-bfd5	1222	20	10	5.57	3.11
PL5	1219	7	4	3.88	1.60
LGd-ip	1176	14	8	11.41	10.07
VIsp2/3	1129	20	10	2.60	0.74
TH	1118	20	10	9.39	6.81
VIsp6a	1079	20	10	3.90	1.78
ACAd5	1070	11	6	4.94	2.20
SSp-bfd6a	1039	20	10	3.81	1.73
VISal6a	994	20	10	3.51	1.63
VIsp4	951	20	10	4.18	1.89
VISam2/3	949	20	10	2.82	1.08
PL6a	904	8	5	4.16	1.89
VISam6a	864	20	10	3.83	1.65
DG-po	837	19	10	5.03	2.64
MOs2/3	818	19	10	3.42	1.34
or	807	20	10	5.02	2.15
SSp-bfd2/3	805	20	10	2.87	1.03
VPL	796	6	4	12.09	9.07
LP	760	11	7	8.39	6.82
VISal4	713	20	10	3.68	1.96
MGd	700	15	9	10.95	8.94
VISal2/3	681	20	10	2.84	1.03
ACAv5	646	8	6	5.20	2.55

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
STR	645	16	10	3.87	1.98
ViSa2/3	642	17	9	3.09	1.36
MOs6a	626	9	5	4.07	1.54
ViSa5	562	12	7	5.62	3.32
MGm	562	12	7	8.24	6.24
SGN	544	13	8	10.31	9.09
ProS	522	8	6	6.29	3.54
ccg	482	12	9	1.42	0.76
ZI	465	10	7	5.98	2.11
POST	438	6	4	6.10	3.13
ViSa6a	433	13	8	3.96	1.81
fp	429	20	10	3.05	1.16
alv	427	20	10	2.40	0.87
MS	389	3	2	7.46	2.79
ViSam4	382	20	10	4.36	2.29
SH	373	7	5	3.63	1.63
ViSrl5	343	7	4	6.05	3.38
ACA6a	334	9	6	4.27	1.87
ACAv2/3	315	7	5	3.54	1.29
ILA5	300	2	2	3.02	0.93
dhc	293	14	9	2.19	1.40
ar	266	12	7	6.73	3.04
IGL	262	9	6	8.28	4.60
PoT	236	5	4	7.48	4.97
ccs	229	19	10	2.59	1.02
fiber tracts	229	20	10	1.29	0.54
ViSrl6a	224	7	4	3.77	2.06
ViSa4	218	12	7	4.75	2.89
CA2	214	18	10	4.98	2.74
fa	204	6	5	1.49	0.62
ViSal6b	202	19	10	4.36	1.97
PL2/3	193	3	3	3.40	1.78
CP	184	1	1	4.80	2.47
ViSrl4	182	5	4	5.73	3.03
RT	181	5	3	11.35	5.10
ViSam6b	172	19	10	3.15	1.64
TTd	169	3	2	3.14	1.26
ViSp6b	168	19	10	3.83	2.31
ACAv1	141	6	3	3.22	1.80
fi	140	6	4	1.58	0.46
SSp-bfd6b	125	17	10	4.43	2.07
SubG	124	1	1	3.68	1.38
root	121	16	10	1.24	0.77

Region label	Units	Files	Subjects	firing_rate mean (Hz)	firing_rate median (Hz)
ViSa6b	121	12	7	3.61	2.02
DP	118	3	2	3.60	1.28
ViSp1	115	14	9	0.97	0.44
IntG	113	4	4	6.43	3.20
ViSrl2/3	109	5	4	5.07	1.68
PIL	90	5	5	5.27	2.80
PL6b	78	3	2	3.95	3.07
LD	75	1	1	11.71	11.76
IG	74	5	3	3.87	1.53
SSp-bfd1	74	15	9	0.87	0.27
LSc	74	2	2	2.72	1.25
bsc	71	17	10	1.46	0.80
PO	70	1	1	9.29	9.02
OLF	70	3	2	2.30	0.96
ACAv6a	58	3	3	4.04	1.77
ViSam1	56	11	7	1.41	0.73
ACA2/3	56	1	1	2.68	0.38
ILA2/3	56	2	1	2.40	1.13
ccb	48	4	3	0.92	0.49
cing	48	9	6	1.42	0.63
scwm	44	9	7	2.62	1.04
PP	42	1	1	5.06	3.60
LSv	38	1	1	7.22	3.16
ViSa1	38	13	9	1.68	1.17
MOs1	38	12	7	1.01	0.55
ILA1	37	1	1	2.02	0.64
HPF	37	7	6	3.42	1.75
VL	36	3	2	3.79	0.87
ViSa1	35	15	9	0.65	0.37
PL1	26	1	1	3.50	1.34
ViSrl6b	24	6	4	3.31	1.74
MB	23	2	2	14.65	10.22
LT	19	2	2	1.40	0.78
em	16	2	1	8.46	5.67
DT	11	2	2	1.18	0.97
ACAv6b	8	2	2	0.56	0.22
POL	8	1	1	1.21	1.25
MOs6b	4	1	1	2.74	1.22
ILA6b	3	1	1	2.73	2.53
SEZ	3	2	2	0.59	0.62
chpl	1	1	1	10.64	10.64

6. Intervals tables (stimuli, trials, epochs)

Each NWB intervals table, with the number of files containing it, total rows, and its columns as stored.

Table	Files	Rows	Columns
40_hz_pulse_train_presentations	20	2,877	start_time, stop_time, condition_num, level, condition, stim_name, duration, name
5_hz_pulse_train_presentations	20	2,877	start_time, stop_time, condition_num, level, condition, stim_name, duration, name
behavior_presentations	20	71,956	start_time, stop_time, duration, image_index, image_name, image_set, omitted, stim_name
post_receptive_field_mapping_presentations	20	38,880	start_time, stop_time, duration, stim_name, contrast, temporal_frequency, spatial_frequency, orientation, pos_x, pos_y
post_surround_suppression_mapping_presentations	20	15,000	start_time, stop_time, duration, stim_name, contrast, temporal_frequency, spatial_frequency, orientation, size, pos_x, pos_y
pre_receptive_field_mapping_presentations	20	38,880	start_time, stop_time, duration, stim_name, contrast, temporal_frequency, spatial_frequency, orientation, pos_x, pos_y
pre_surround_suppression_mapping_presentations	20	15,000	start_time, stop_time, duration, stim_name, contrast, temporal_frequency, spatial_frequency, orientation, size, pos_x, pos_y
raised_cosine_presentations	20	2,878	start_time, stop_time, condition_num, level, condition, stim_name, duration, name
spontaneous_presentations	20	101,976	start_time, stop_time, duration, stim_name

7. Column and signal descriptions from the NWB files

Quoted verbatim from sub-752309_ses-ecephys-752309-2025-02-11-14-00-40_ecephys.nwb; all files in the Dandiset share this structure.

Location in NWB file	Name	Description (as written in the NWB file)
electrodes	(table)	metadata about extracellular electrodes
electrodes	location	the location of channel within the subject e.g. brain region
electrodes	group	a reference to the ElectrodeGroup this electrode is a part of
electrodes	group_name	the name of the ElectrodeGroup this electrode is a part of
electrodes	channel_name	unique channel reference
electrodes	rel_x	rel_x
electrodes	rel_y	rel_y
electrodes	inter_sample_shift	no description
electrodes	x	ccf x coordinate
electrodes	y	ccf y coordinate
electrodes	z	ccf z coordinate
units	(table)	Units from Kilosort4
units	spike_times	the spike times for each unit
units	electrodes	the electrodes that each spike unit came from
units	waveform_mean	the spike waveform mean for each spike unit
units	waveform_sd	the spike waveform standard deviation for each spike unit
units	unit_name	Unique reference for each unit.
units	amplitude_median	No description.
units	depth	No description.
units	isi_violations_ratio	No description.
units	exp_decay	No description.
units	velocity_below	No description.
units	nn_hit_rate	No description.
units	amplitude	No description.
units	peak_to_valley	The duration between the negative and the positive peaks computed on the maximum channel.
units	rp_contamination	No description.
units	d_prime	No description.
units	drift_ptp	No description.
units	num_spikes	No description.
units	snr	The signal-to-noise ratio of the unit.
units	amplitude_cv_median	No description.
units	nn_miss_rate	No description.
units	original_cluster_id	No description.
units	silhouette	No description.
units	velocity_above	No description.
units	ks_unit_id	No description.
units	sync_spike_2	No description.
units	device_name	No description.
units	estimated_x	No description.
units	peak_trough_ratio	No description.

Location in NWB file	Name	Description (as written in the NWB file)
units	sliding_rp_violation	No description.
units	isi_violations_count	No description.
units	isolation_distance	No description.
units	repolarization_slope	No description.
units	I_ratio	No description.
units	spread	No description.
units	decoder_probability	No description.
units	rp_violations	No description.
units	drift_mad	No description.
units	sync_spike_8	No description.
units	num_negative_peaks	No description.
units	presence_ratio	No description.
units	amplitude_cv_range	No description.
units	half_width	No description.
units	estimated_z	No description.
units	num_positive_peaks	No description.
units	default_qc	No description.
units	firing_rate	Number of spikes per unit of time.
units	shank	No description.
units	recovery_slope	No description.
units	decoder_label	No description.
units	drift_std	No description.
units	amplitude_cutoff	No description.
units	estimated_y	No description.
units	firing_range	No description.
units	sync_spike_4	No description.
intervals/40 hz pulse train_presentations	(table)	Presentation times and stimuli details for '40 hz pulse train' stimuli. Note: image_name references control_description in stimulus/templates
intervals/40 hz pulse train_presentations	start_time	Start time of epoch, in seconds
intervals/40 hz pulse train_presentations	stop_time	Stop time of epoch, in seconds
intervals/40 hz pulse train_presentations	condition_num	Description of condition_num
intervals/40 hz pulse train_presentations	level	Description of level
intervals/40 hz pulse train_presentations	condition	Description of condition
intervals/40 hz pulse train_presentations	stim_name	Description of stim_name
intervals/40 hz pulse train_presentations	duration	Description of duration
intervals/40 hz pulse train_presentations	timeseries	index into a TimeSeries object
intervals/5 hz pulse train_presentations	(table)	Presentation times and stimuli details for '5 hz pulse train' stimuli. Note: image_name references control_description in stimulus/templates

Location in NWB file	Name	Description (as written in the NWB file)
intervals/5 hz pulse train_presentations	start_time	Start time of epoch, in seconds
intervals/5 hz pulse train_presentations	stop_time	Stop time of epoch, in seconds
intervals/5 hz pulse train_presentations	condition_num	Description of condition_num
intervals/5 hz pulse train_presentations	level	Description of level
intervals/5 hz pulse train_presentations	condition	Description of condition
intervals/5 hz pulse train_presentations	stim_name	Description of stim_name
intervals/5 hz pulse train_presentations	duration	Description of duration
intervals/5 hz pulse train_presentations	timeseries	index into a TimeSeries object
intervals/behavior_presentations	(table)	Presentation times and stimuli details for 'behavior' stimuli. Note: image_name references control_description in stimulus/templates
intervals/behavior_presentations	start_time	Start time of epoch, in seconds
intervals/behavior_presentations	stop_time	Stop time of epoch, in seconds
intervals/behavior_presentations	duration	Description of duration
intervals/behavior_presentations	image_name	Description of image_name
intervals/behavior_presentations	image_set	Description of image_set
intervals/behavior_presentations	omitted	Description of omitted
intervals/behavior_presentations	stim_name	Description of stim_name
intervals/behavior_presentations	timeseries	index into a TimeSeries object
intervals/injection_times	(table)	Intervals from ../data/interval_metadata/injection_times.csv
intervals/injection_times	start_time	Start time of epoch, in seconds
intervals/injection_times	stop_time	Stop time of epoch, in seconds
intervals/injection_times	lims_session_id	lims_session_id column from CSV
intervals/injection_times	session_id	session_id column from CSV
intervals/injection_times	injection_type	injection_type column from CSV
intervals/post_receptive_field_mapping_presentations	(table)	Presentation times and stimuli details for 'post_receptive_field_mapping' stimuli. Note: image_name references control_description in stimulus/templates
intervals/post_receptive_field_mapping_presentations	start_time	Start time of epoch, in seconds
intervals/post_receptive_field_mapping_presentations	stop_time	Stop time of epoch, in seconds
intervals/post_receptive_field_mapping_presentations	duration	Description of duration
intervals/post_receptive_field_mapping_presentations	stim_name	Description of stim_name
intervals/post_receptive_field_mapping_presentations	contrast	Description of contrast
intervals/post_receptive_field_mapping_presentations	temporal_frequency	Description of temporal_frequency
intervals/post_receptive_field_mapping_presentations	spatial_frequency	Description of spatial_frequency
intervals/post_receptive_field_mapping_presentations	orientation	Description of orientation
intervals/post_receptive_field_mapping_presentations	pos_x	Description of pos_x
intervals/post_receptive_field_mapping_presentations	pos_y	Description of pos_y

Location in NWB file	Name	Description (as written in the NWB file)
intervals/post_receptive_field_mapping_presentations	timeseries	index into a TimeSeries object
intervals/post_surround_suppression_mapping_presentations	(table)	Presentation times and stimuli details for 'post_surround_suppression_mapping' stimuli. Note: image_name references control_description in stimulus/templates
intervals/post_surround_suppression_mapping_presentations	start_time	Start time of epoch, in seconds
intervals/post_surround_suppression_mapping_presentations	stop_time	Stop time of epoch, in seconds
intervals/post_surround_suppression_mapping_presentations	duration	Description of duration
intervals/post_surround_suppression_mapping_presentations	stim_name	Description of stim_name
intervals/post_surround_suppression_mapping_presentations	contrast	Description of contrast
intervals/post_surround_suppression_mapping_presentations	temporal_frequency	Description of temporal_frequency
intervals/post_surround_suppression_mapping_presentations	spatial_frequency	Description of spatial_frequency
intervals/post_surround_suppression_mapping_presentations	orientation	Description of orientation
intervals/post_surround_suppression_mapping_presentations	size	Description of size
intervals/post_surround_suppression_mapping_presentations	pos_x	Description of pos_x
intervals/post_surround_suppression_mapping_presentations	pos_y	Description of pos_y
intervals/post_surround_suppression_mapping_presentations	timeseries	index into a TimeSeries object
intervals/pre_receptive_field_mapping_presentations	(table)	Presentation times and stimuli details for 'pre_receptive_field_mapping' stimuli. Note: image_name references control_description in stimulus/templates
intervals/pre_receptive_field_mapping_presentations	start_time	Start time of epoch, in seconds
intervals/pre_receptive_field_mapping_presentations	stop_time	Stop time of epoch, in seconds
intervals/pre_receptive_field_mapping_presentations	duration	Description of duration
intervals/pre_receptive_field_mapping_presentations	stim_name	Description of stim_name
intervals/pre_receptive_field_mapping_presentations	contrast	Description of contrast
intervals/pre_receptive_field_mapping_presentations	temporal_frequency	Description of temporal_frequency
intervals/pre_receptive_field_mapping_presentations	spatial_frequency	Description of spatial_frequency
intervals/pre_receptive_field_mapping_presentations	orientation	Description of orientation
intervals/pre_receptive_field_mapping_presentations	pos_x	Description of pos_x

Location in NWB file	Name	Description (as written in the NWB file)
intervals/pre_receptive_field_mapping_presentations	pos_y	Description of pos_y
intervals/pre_receptive_field_mapping_presentations	timeseries	index into a TimeSeries object
intervals/pre_surround_suppression_mapping_presentations	(table)	Presentation times and stimuli details for 'pre_surround_suppression_mapping' stimuli. Note: image_name references control_description in stimulus/templates
intervals/pre_surround_suppression_mapping_presentations	start_time	Start time of epoch, in seconds
intervals/pre_surround_suppression_mapping_presentations	stop_time	Stop time of epoch, in seconds
intervals/pre_surround_suppression_mapping_presentations	duration	Description of duration
intervals/pre_surround_suppression_mapping_presentations	stim_name	Description of stim_name
intervals/pre_surround_suppression_mapping_presentations	contrast	Description of contrast
intervals/pre_surround_suppression_mapping_presentations	temporal_frequency	Description of temporal_frequency
intervals/pre_surround_suppression_mapping_presentations	spatial_frequency	Description of spatial_frequency
intervals/pre_surround_suppression_mapping_presentations	orientation	Description of orientation
intervals/pre_surround_suppression_mapping_presentations	size	Description of size
intervals/pre_surround_suppression_mapping_presentations	pos_x	Description of pos_x
intervals/pre_surround_suppression_mapping_presentations	pos_y	Description of pos_y
intervals/pre_surround_suppression_mapping_presentations	timeseries	index into a TimeSeries object
intervals/raised_cosine_presentations	(table)	Presentation times and stimuli details for 'raised_cosine' stimuli. Note: image_name references control_description in stimulus/templates
intervals/raised_cosine_presentations	start_time	Start time of epoch, in seconds
intervals/raised_cosine_presentations	stop_time	Stop time of epoch, in seconds
intervals/raised_cosine_presentations	condition_num	Description of condition_num
intervals/raised_cosine_presentations	level	Description of level
intervals/raised_cosine_presentations	condition	Description of condition
intervals/raised_cosine_presentations	stim_name	Description of stim_name
intervals/raised_cosine_presentations	duration	Description of duration
intervals/raised_cosine_presentations	timeseries	index into a TimeSeries object
intervals/spontaneous_presentations	(table)	Presentation times and stimuli details for 'spontaneous' stimuli. Note: image_name references control_description in stimulus/templates
intervals/spontaneous_presentations	start_time	Start time of epoch, in seconds
intervals/spontaneous_presentations	stop_time	Stop time of epoch, in seconds
intervals/spontaneous_presentations	duration	Description of duration
intervals/spontaneous_presentations	stim_name	Description of stim_name
intervals/spontaneous_presentations	timeseries	index into a TimeSeries object
acquisition	autorewarded	Boolean flag indicating whether reward was auto-delivered
acquisition	licks	Lick events

Location in NWB file	Name	Description (as written in the NWB file)
acquisition	raw_running_wheel_rotation	no description
acquisition	reward_volume	Volume of reward delivered
acquisition	running_wheel_signal_voltage	no description
acquisition	running_wheel_supply_voltage	no description
processing/ecephys	(module)	Intermediate data from extracellular electrophysiology recordings, e.g., LFP.
processing/ecephys/LFP	ElectricalSeriesProbeA-LFP	LFP voltage traces from ProbeA
processing/ecephys/LFP	ElectricalSeriesProbeB-LFP	LFP voltage traces from ProbeB
processing/ecephys/LFP	ElectricalSeriesProbeC-LFP	LFP voltage traces from ProbeC
processing/ecephys/LFP	ElectricalSeriesProbeD-LFP	LFP voltage traces from ProbeD
processing/ecephys/LFP	ElectricalSeriesProbeE-LFP	LFP voltage traces from ProbeE
processing/ecephys/LFP	ElectricalSeriesProbeF-LFP	LFP voltage traces from ProbeF
processing/eye_tracking	(module)	eye tracking data
processing/eye_tracking	likely_blink_times	no description
processing/eye_tracking/corneal_reflection	(table)	
processing/eye_tracking/corneal_reflection	reference_frame	no description
processing/eye_tracking/corneal_reflection	data_x	no description
processing/eye_tracking/corneal_reflection	data_y	no description
processing/eye_tracking/corneal_reflection	area	no description
processing/eye_tracking/corneal_reflection	area_raw	no description
processing/eye_tracking/corneal_reflection	width	no description
processing/eye_tracking/corneal_reflection	height	no description
processing/eye_tracking/corneal_reflection	angle	no description
processing/eye_tracking/corneal_reflection	timestamps	no description
processing/eye_tracking/ellipse	(table)	
processing/eye_tracking/ellipse	reference_frame	no description
processing/eye_tracking/ellipse	data_x	no description
processing/eye_tracking/ellipse	data_y	no description
processing/eye_tracking/ellipse	area	no description
processing/eye_tracking/ellipse	area_raw	no description
processing/eye_tracking/ellipse	width	no description
processing/eye_tracking/ellipse	height	no description
processing/eye_tracking/ellipse	angle	no description
processing/eye_tracking/ellipse	timestamps	no description
processing/eye_tracking/pupil	(table)	
processing/eye_tracking/pupil	reference_frame	no description

Location in NWB file	Name	Description (as written in the NWB file)
processing/eye_tracking/pupil	data_x	no description
processing/eye_tracking/pupil	data_y	no description
processing/eye_tracking/pupil	area	no description
processing/eye_tracking/pupil	area_raw	no description
processing/eye_tracking/pupil	width	no description
processing/eye_tracking/pupil	height	no description
processing/eye_tracking/pupil	angle	no description
processing/eye_tracking/pupil	timestamps	no description
processing/running	(module)	running speed data
processing/running	running_speed	no description
processing/running	running_wheel_rotation	no description
processing/stimulus	(module)	Stimulus Times processing
processing/stimulus	timestamps	no description

8. Exported files

File	Rows	Columns
tables/sessions.csv	20	session_id, asset_path, size_gb, subject_id, species, sex, age, genotype, strain, session_start_time, session_description, institution, lab, n_units, n_electrodes, electrode_groups, region_source, regions, intervals, acquisition, processing
tables/units.parquet	82,742	session_id, unit_id, unit_row, region, unit_name, amplitude_median, depth, isi_violations_ratio, exp_decay, velocity_below, nn_hit_rate, amplitude, peak_to_valley, rp_contamination, d_prime, drift_ptp, num_spikes, snr, amplitude_cv_median, nn_miss_rate, original_cluster_id, silhouette, velocity_above, ks_unit_id, sync_spike_2, device_name, estimated_x, peak_trough_ratio, sliding_rp_violation, isi_violations_count, isolation_distance, repolarization_slope, l_ratio, spread, decoder_probability, rp_violations, drift_mad, sync_spike_8, num_negative_peaks, presence_ratio, amplitude_cv_range, half_width, estimated_z, num_positive_peaks, default_qc, firing_rate, shank, recovery_slope, decoder_label, drift_std, amplitude_cutoff, estimated_y, firing_range, sync_spike_4, subject_id
tables/region_summary.csv	124	region, n_units, n_sessions, n_subjects, firing_rate_mean_hz, firing_rate_median_hz
tables/intervals_summary.csv	9	table, n_sessions, n_rows, columns
intervals/<table>.parquet	one per intervals table	session_id + the table's columns
tables/dataset_metadata.json, nwb_file_metadata.json, nwb_descriptions.csv		published metadata quoted in sections 1 and 7